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Cohort : Jensen Huang

Roll No : 150096724023

Subject : AWS

Problem Statement No : 90

Problem Statement Title : 90. eSportArena Competitive Gaming Cloud

Sem : 4 , **Sprint :** 02

Industry: Esports & Competitive Gaming

Github Repo Link :

<https://github.com/RaginiSingh2024/eSportArena-Competitive-Gaming-Cloud>

Documentation Link :

<https://docs.google.com/document/d/1Dcg3mRjWg-Ra0vMbqVI3JZPwxyE2xVwinEvOqA5N-fo/edit?usp=sharing>

1. Abstract

eSportArena is a cloud-based gaming platform infrastructure deployed on Amazon Web Services (AWS). The project demonstrates the implementation of core AWS services including VPC, Public Subnets, Security Groups, EC2 Instance, Ubuntu Server, and Nginx Web Server.

The objective is to create a scalable, secure, and highly available cloud environment capable of hosting web applications. The website is successfully deployed on an EC2 instance and accessed through a public IP address.

2. Objectives

- Deploy a website on AWS Cloud.
- Configure networking using VPC and Subnets.
- Launch and manage an EC2 instance.
- Install Ubuntu and Nginx Web Server.
- Understand cloud infrastructure deployment.

- Demonstrate secure and scalable cloud architecture.

3. Technologies Used

Technology	Purpose
AWS EC2	Virtual Server
AWS VPC	Network Isolation
Public subnet	Internet Access
Security Group	Firewall Rules
Ubuntu 24.04	Operating System
Nginx	Web Server
Draw.io	Architecture Diagram
HTML/CSS	Landing page

4. Folder Structure

eSportArena-AWS-Project/

- | |— README.md
- |— Architecture.md
- |— Project_Report.docx
- |— Project_Presentation.pptx
- |— website/
 - | |— index.html
 - | |— styles.css (agar CSS alag file me hai)
- | |— diagrams/
 - | |— aws_architecture.drawio
 - | |— aws_architecture.png
 - | |— aws_architecture.pdf
- | |— screenshots/

- | |— 01_vpc_created.png
- | |— 02_public_subnets.png
- | |— 03_security_group.png
- | |— 04_ec2_running.png
- | |— 05_ubuntu_terminal.png
- | |— 06_nginx_installed.png
- | |— 07_website_live.png
- | |— 08_architecture_diagram.png
- | |— deployment/
- | |— deployment_steps.md
- | |— nginx_commands.txt
- | |— aws_configuration.txt |
- |— docs/
- |— Abstract.md
- |— Objectives.md
- |— Results.md

|— Future_Scope.md |— [Conclusion.md](#)

4. Architecture Overview

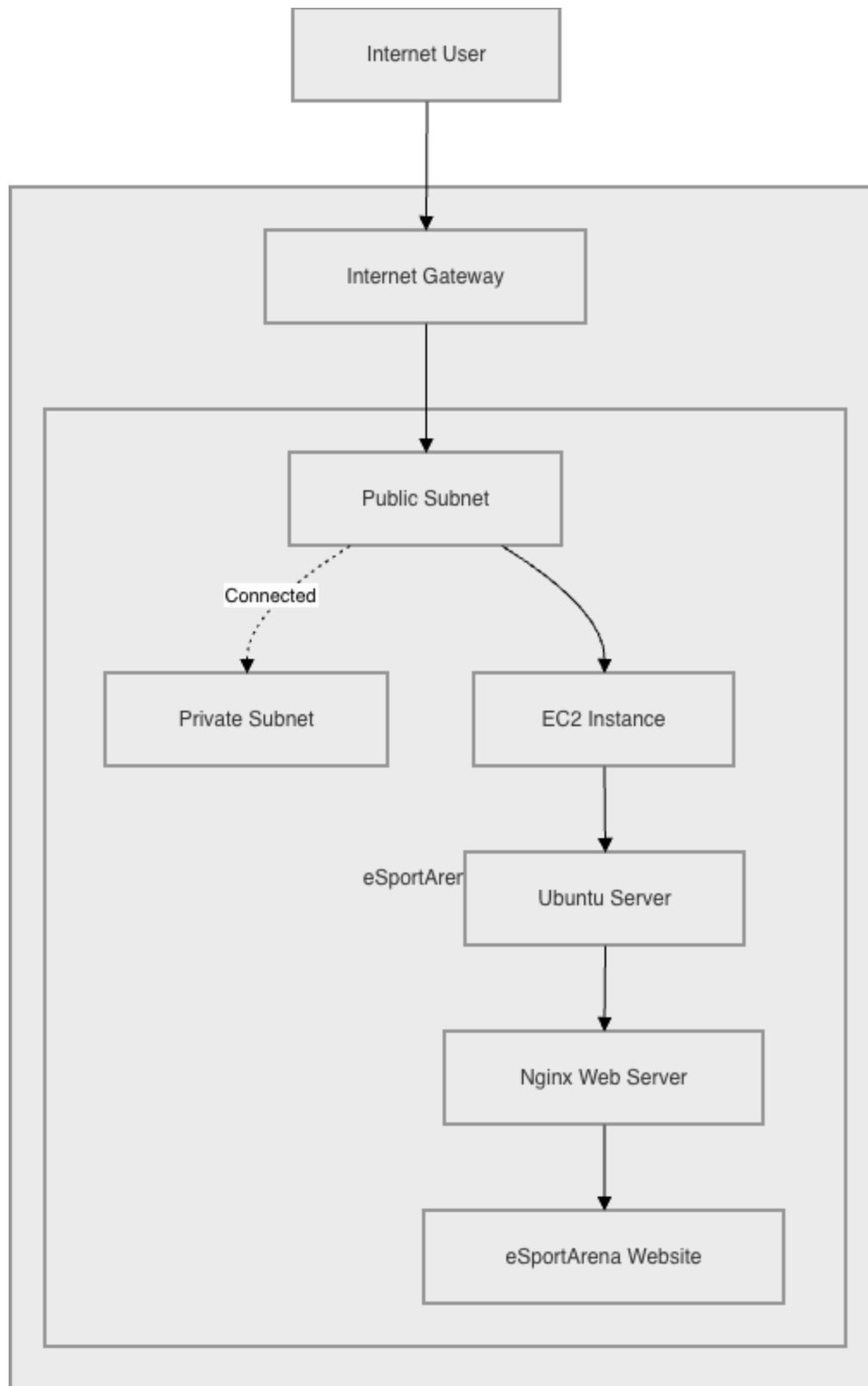


Figure 1: AWS Cloud Architecture of eSportArena Platform

5. Implementation Steps

Step 1: Create VPC

Created a Virtual Private Cloud to isolate network resources.

Step 2: Create Public Subnets

Created two public subnets for resource deployment.

Step 3: Configure Security Groups

Allowed SSH (Port 22) and HTTP (Port 80) traffic.

Step 4: Launch EC2 Instance

Created an Ubuntu-based EC2 instance.

Step 5: Install Nginx

Installed and enabled Nginx web server.

Step 6: Deploy Website

Uploaded a custom HTML page and hosted it using Nginx.

Step 7: Test Deployment

Verified successful deployment using the public IP address.

Step 8: Configure MySQL Database

Installed and configured MySQL Database Server on the EC2 instance.
Verified database service status and connectivity.

Step 9: Linux User Management

Created a new Linux user and configured user access permissions for system administration tasks.

Step 10: Permission Management

Assigned appropriate file and directory permissions to ensure secure access control.

Step 11: Configure Cron Jobs

Created and tested cron jobs for automated task scheduling and system administration activities.

Step 12: Install Docker

Installed Docker Engine on the Ubuntu EC2 instance and verified successful installation.

Step 13: Deploy Docker Container

Created and executed a Docker container to demonstrate containerized application deployment.

Step 14: Create Shell Script Automation

Developed and executed shell scripts to automate administrative and deployment-related tasks.

Step 15: Create IAM User

Created an IAM user with appropriate permissions to demonstrate AWS Identity and Access Management concepts.

Step 16: Create Amazon S3 Bucket

Created and configured an Amazon S3 bucket for cloud storage demonstration.

Step 17: Configure CloudWatch Monitoring

Accessed CloudWatch Metrics and monitored EC2 instance performance and resource utilization.

Step 18: Create AWS Architecture Diagram

Designed the complete AWS architecture diagram using Draw.io, illustrating VPC, Subnets, Security Groups, EC2 Instance, Ubuntu Server, Nginx Web Server, and Website Deployment.

Step 19: Prepare Documentation

Created README.md, Architecture.md, deployment documentation, architecture diagrams, screenshots, and final project report.

Step 20: Upload Project to GitHub

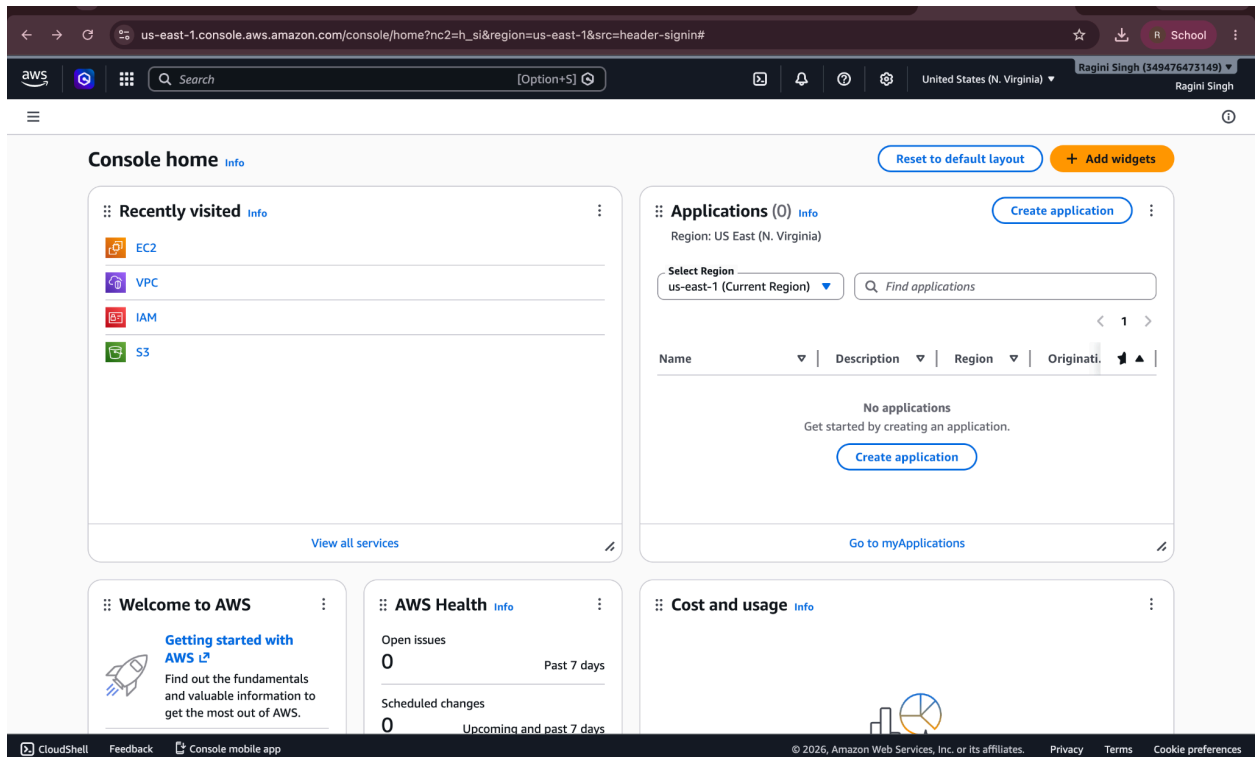
Created a GitHub repository and uploaded source code, architecture diagrams, documentation files, and deployment screenshots.

Step 21: Final Verification

Verified AWS resources, website deployment, monitoring configuration, documentation, and repository structure before project submission.

6. Screenshots

1. Aws_login_page.png



2. SS-02-VPC-Created.png

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateVpcWizard:

aws

Search

[Option+S]

United States (N. Virginia)

Ragini Singh (349476473149)

Ragini Singh

VPC > Your VPCs > Create VPC > Create VPC resources

Create VPC workflow

Success

Details

Create VPC: vpc-020690e63f9c8b111

Enable DNS hostnames

Enable DNS resolution

Verifying VPC creation: vpc-020690e63f9c8b111

Create subnet: subnet-008d05299ebd81ffb

Create subnet: subnet-0ff3e820f0753e077

Create internet gateway: igw-02c427bb185d2472b

Attach internet gateway to the VPC

Create route table: rtb-063727948a67a1ee2

Create route

Associate route table

Create route table: rtb-0e4165dd11b6e8aa4

Associate route table

Verifying route table creation

Related VPC resources and services

Discover additional resources you can launch within your VPC to build your network architecture. From security components to connectivity services, explore the building blocks available for your infrastructure.

Client VPN

Networking

AWS Client VPN is a managed client-based VPN service that enables you to securely access AWS resources and resources in your on-premises network.

Resources

Let us know what you would like to create in your VPC

e.g., Instances, Network ACL, Transit gateway.

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3. SS-02-VPC-List.png

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#vpcs:

aws [Search] [Option+S] United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

VPC > Your VPCs

VPC dashboard < AWS Global View ⓘ Filter by VPC: ▼

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

- Getting started
- Endpoints

Your VPCs VPC encryption controls

Your VPCs (3) Info Last updated less than a minute ago Actions ▼ Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...
☐	jensen_VPC	vpc-0b080cd0e0b0b1635	Available	-	-	Off
☐	-	vpc-0e61863fad3218680	Available	-	-	Off
☐	eSportArena-vpc	vpc-020690e63f9c8b111	Available	-	-	Off

Select a VPC above

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4. SS-03-Resource-Map.png

aws [Search] [Option+S] United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

VPC > Your VPCs > vpc-020690e63f9c8b111

VPC dashboard

AWS Global View [🌐]

Filter by VPC: [v]

Virtual private cloud

Security

PrivateLink and Lattice

DNS firewall

Network Firewall

Details

VPC ID: vpc-020690e63f9c8b111

State: Available

DNS resolution: Enabled

Main network ACL: acl-0097d3176257b8fdc

IPv6 CIDR (Network border group): -

Encryption control ID: -

Tenancy: default

Default VPC: No

Network Address Usage metrics: Disabled

Encryption control mode: -

Block Public Access: Off

DHCP option set: dopt-09c7f60479a105b63

IPv4 CIDR: 10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups: -

DNS hostnames: Enabled

Main route table: rtb-0b541591be6befb0b

IPv6 pool: -

Owner ID: 349476473149

Resource map | CIDRs | Flow logs | Tags | Integrations

Resource map

VPC: Your AWS virtual network

Subnets (2): Subnets within this VPC

Route tables (3): Route network traffic to resources

us-east-1a

eSportArena-subnet-public1-us-east-1a

eSportArena-subnet-private1-us-east-1a

eSportArena-rtb-public

eSportArena-rtb-private1-us-east-1a

rtb-0b541591be6befb0b

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5. SS-03-Subnets.png

aws [Search] [Option+S] United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

VPC > Subnets

VPC dashboard

AWS Global View [🌐]

Filter by VPC: [v]

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Subnets (9)

Find subnets by attribute or tag

Last updated less than a minute ago

Actions [⌵] Create subnet

	Name	Subnet ID	State	VPC	Block Public...	IPv4 C
☐	-	subnet-00018395da4d2aea8	Available	vpc-0e61863fad3218680	Off	172.31
☐	-	subnet-08f2b2409e39f7490	Available	vpc-0e61863fad3218680	Off	172.31
☐	eSportArena-subnet-public1-us-east-1a	subnet-008d05299ebd81ffb	Available	vpc-020690e63f9c8b111 eSp...	Off	10.0.0
☐	-	subnet-04ad2f5ecd9a3f171	Available	vpc-0e61863fad3218680	Off	172.31
☐	-	subnet-07880e1cd3d341aa9	Available	vpc-0e61863fad3218680	Off	172.31
☐	-	subnet-0bfcd0d7d5dd95a78	Available	vpc-0e61863fad3218680	Off	172.31
☐	-	subnet-0734dc1c400642f6d	Available	vpc-0e61863fad3218680	Off	172.31
☐	eSportArena-subnet-private1-us-east-1a	subnet-0ff3e820f0753e077	Available	vpc-020690e63f9c8b111 eSp...	Off	10.0.1
☐	jensen_subnet	subnet-0281a306d59a5410e	Available	vpc-0b080cd0e0b0b1635 jens...	Off	10.0.1

Select a subnet

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6. SS-04-EC2-Configuration.png

aws [Search] [Option+S] United States (N. Virginia) Ragini Singh (\$49476473149) Ragini Singh

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices. [Take a walkthrough](#) [Do not show me this message again.](#)

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

eSportArena-WebServer [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-0152204c1a187337c

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

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7. SS-05-KeyPair-Created.png

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EC2 > Instances > Launch an instance

Instance type [Info](#) [Get advice](#)

Instance type

t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

[Compare instance types](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

eSportArena-Key [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)

vpc-0e61863fad3218680

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-091138d0f0d41ff90

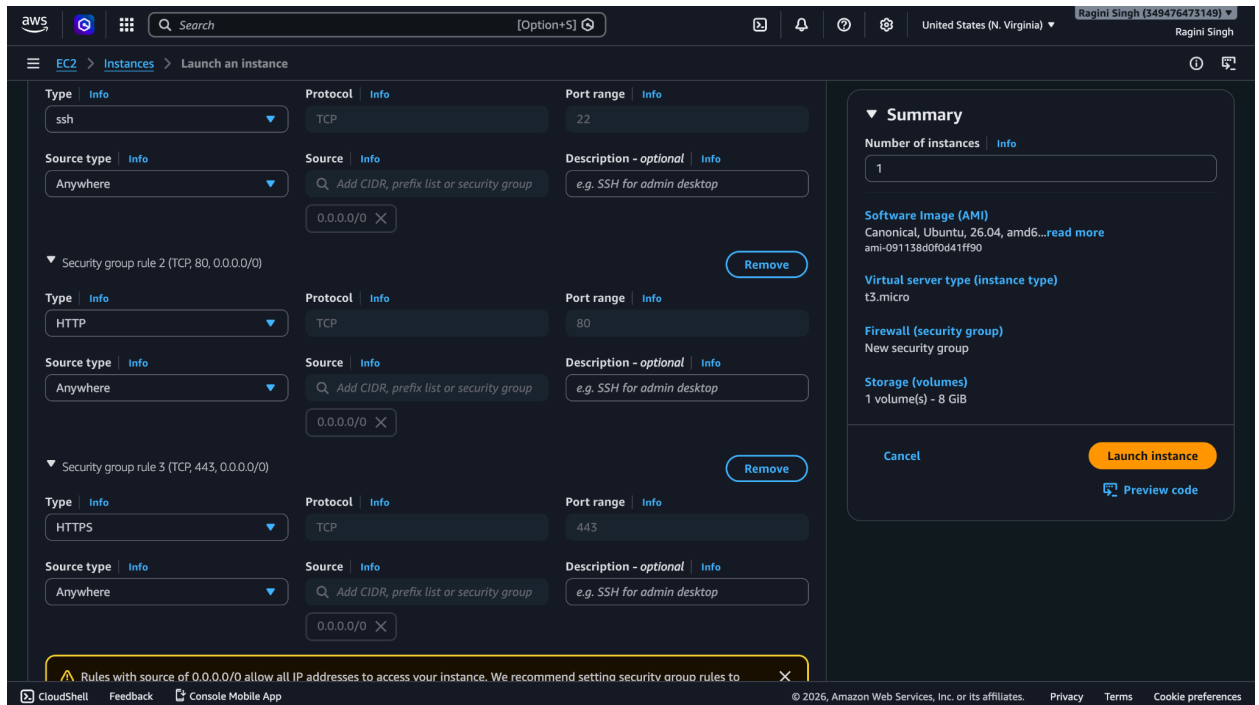
Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

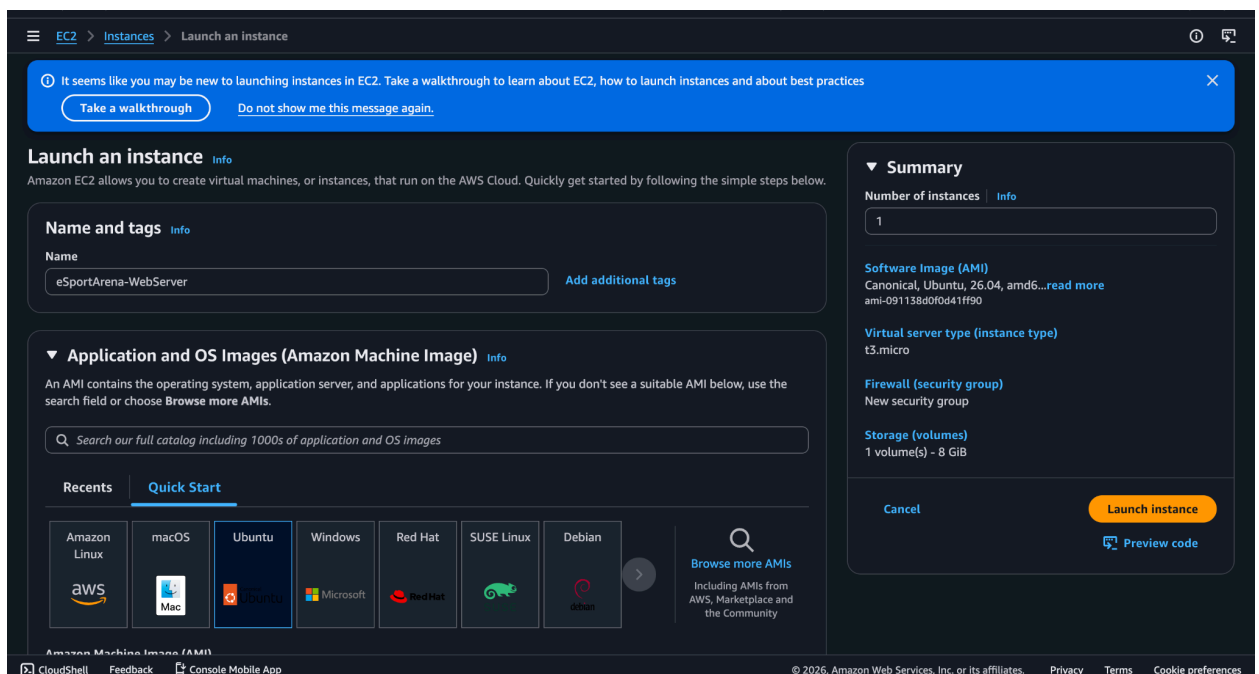
Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

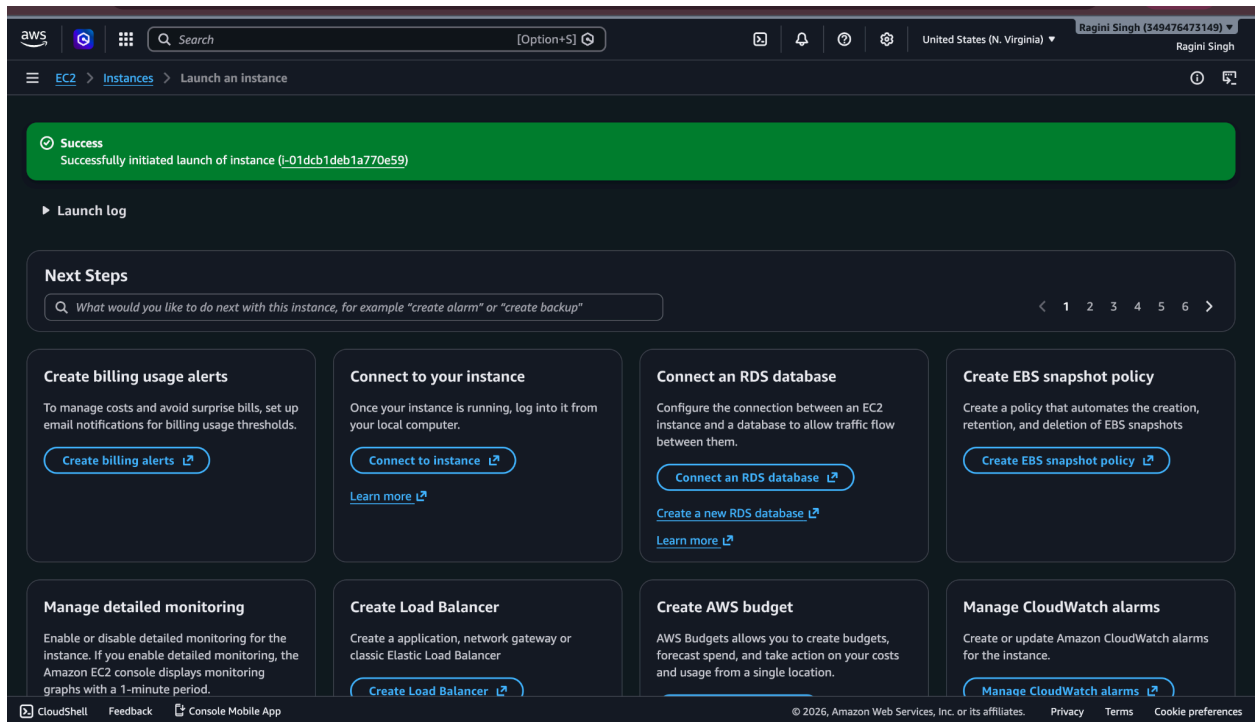
8. SS-07-Security-Group-Rules.png



9. SS-08-EC2-Launch-Configuration.png



10. SS-09-EC2-Running-Instance.png



11. SS-09-EC2-Started-Again.png

Cloud Computing Services - x Instances | EC2 | us-east-1 x Inbox (251) - 2024.raginis@... x + Ask Gemini

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:

Search [Option+S] United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images
 - AMIs
 - AMI Catalog
- Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager
- Network & Security
 - Security Groups

Instances (1/13) Info

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	frontend-server	i-0113b12371b4a8d31	Stopped	t3.micro	-	us-east-1f	-
☐	aws practice	i-09c1061a4003baa36	Stopped	t3.micro	-	us-east-1f	-
☐	awstask	i-07ce58749594c3d01	Stopped	t3.micro	-	us-east-1f	-
☐	backend-server	i-05fbb4bf1b5fa7859	Stopped	t3.micro	-	us-east-1f	-
☒	eSportArena-...	i-01dcb1deb1a770e59	Running	t3.micro	Initializing	us-east-1a	ec2-13-218-119-220.co.
☐	docker	i-0c9b0a7a9c8fe6c78	Stopped	t3.micro	-	us-east-1d	-

i-01dcb1deb1a770e59 (eSportArena-WebServer)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-01dcb1deb1a770e59	Public IPv4 address 13.218.119.220 open address	Private IPv4 addresses 10.0.6.25
IPv6 address -	Instance state Running	Public DNS ec2-13-218-119-220.compute-1.amazonaws.com open address
Hostname type ip-10a6-625-220-13.ec2.internal	Private IP DNS name (IPv4 only) ip-10a6-625-220-13.ec2.internal	

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12. SS-10-EC2-Terminal-Connected.png

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01dcb1deb1a770e59&osUser=ubuntu...

aws

Search

[Option+S]

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Ragini Singh (349476473149)

Ragini Singh

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

* Documentation: <https://docs.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Fri Jun 12 08:59:22 UTC 2026

System load: 0.11

Temperature: -273.1 C

Usage of /: 30.4% of 6.61GB

Processes: 119

Memory usage: 26%

Users logged in: 0

Swap usage: 0%

IPv4 address for ens5: 10.0.6.25

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-6-25:~\$

i-01dcb1deb1a770e59 (eSportArena-WebServer)

PublicIPs: 44.195.91.150 PrivateIPs: 10.0.6.25

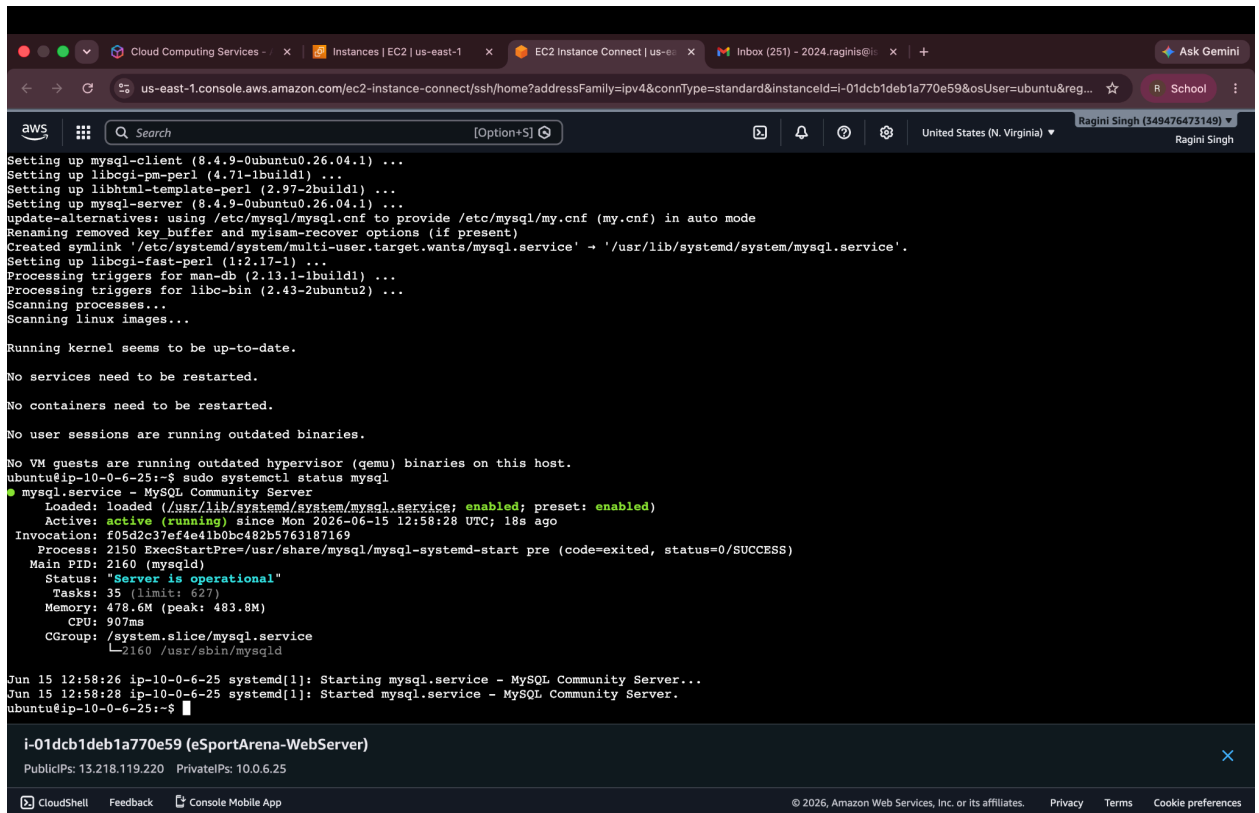
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13. SS-10-MySQL-Running.png



```
Setting up mysql-client (8.4.9-0ubuntu0.26.04.1) ...
Setting up libbgi-pm-perl (4.71-1build1) ...
Setting up libhtml-template-perl (2.97-2build1) ...
Setting up mysql-server (8.4.9-0ubuntu0.26.04.1) ...
update-alternatives: using /etc/mysql/mysql.cnf to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Renaming removed key buffer and myisam-recover options (if present)
Created symlink '/etc/systemd/system/multi-user.target.wants/mysql.service' -> '/usr/lib/systemd/system/mysql.service'.
Setting up libbgi-fast-perl (1:2.17-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-15 12:58:28 UTC; 18s ago
     Invocation: f05d2c37ef4e41b0bc482b5763187169
   Process: 2150 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 2160 (mysqld)
      Status: "Server is operational"
        Tasks: 35 (limit: 627)
       Memory: 478.6M (peak: 483.8M)
          CPU: 907ms
    CGroup: /system.slice/mysql.service
            └─2160 /usr/sbin/mysqld

Jun 15 12:58:26 ip-10-0-6-25 systemd[1]: Starting mysql.service - MySQL Community Server...
Jun 15 12:58:28 ip-10-0-6-25 systemd[1]: Started mysql.service - MySQL Community Server.
ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer)

PublicIPs: 13.218.119.220 PrivateIPs: 10.0.6.25

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14. SS-11-Linux-User-Created.png

```
Cloud Computing Services - x Instances | EC2 | us-east-1 x EC2 Instance Connect | us-e... x Inbox (251) - 2024.raginis@... x + Ask Gemini
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01dcb1deb1a770e59&osUser=ubuntu&reg... School
aws Search [Option+S] United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-15 12:58:28 UTC; 18s ago
 Invocation: f05d2c37ef4e41b0bc482b5763187169
   Process: 2150 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
  Main PID: 2160 (mysqld)
    Status: "Server is operational"
     Tasks: 35 (limit: 627)
  Memory: 478.6M (peak: 483.8M)
     CPU: 907ms
    CGroup: /system.slice/mysql.service
            └─2160 /usr/sbin/mysqld

Jun 15 12:58:26 ip-10-0-6-25 systemd[1]: Starting mysql.service - MySQL Community Server...
Jun 15 12:58:28 ip-10-0-6-25 systemd[1]: Started mysql.service - MySQL Community Server.
ubuntu@ip-10-0-6-25:~$ sudo adduser esportadmin
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for esportadmin
Enter the new value, or press ENTER for the default
  Full Name []: ragini singh
    Room Number []: 01
      Work Phone []:
    Home Phone []:
        Other []:
Is the information correct? [Y/n] y
ubuntu@ip-10-0-6-25:~$ id esportadmin
uid=1001(esportadmin) gid=1001(esportadmin) groups=1001(esportadmin),100(users)
ubuntu@ip-10-0-6-25:~$

i-01dcb1deb1a770e59 (eSportArena-WebServer)
PublicIPs: 13.218.119.220 PrivateIPs: 10.0.6.25
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```

15. SS-11-Nginx-Service-Running.png

```
aws
Search [Option+S]
United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.4) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
sudo systemctl start nginx
Processing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl start nginx
ubuntu@ip-10-0-6-25:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-6-25:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-12 09:00:38 UTC; 34s ago
     Invocation: 5b5c8b8c3033440db407f123b720aa9a
       Docs: man:nginx(8)
    Main PID: 1998 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 39ms
    CGroup: /system.slice/nginx.service
            └─1998 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─2001 "nginx: worker process"
                 └─2002 "nginx: worker process"

Jun 12 09:00:38 ip-10-0-6-25 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jun 12 09:00:38 ip-10-0-6-25 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer)

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15. SS-11-Nginx-Service-Running.png

```
aws
Search [Option+S]
United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.4) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
sudo systemctl start nginxProcessing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl start nginx
ubuntu@ip-10-0-6-25:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-6-25:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-12 09:00:38 UTC; 34s ago
     Invocation: 5b5c8b8c3033440db407f123b720aa9a
       Docs: man:nginx(8)
    Main PID: 1998 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 39ms
    CGroup: /system.slice/nginx.service
            └─1998 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─2001 "nginx: worker process"
                 └─2002 "nginx: worker process"

Jun 12 09:00:38 ip-10-0-6-25 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jun 12 09:00:38 ip-10-0-6-25 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer)

Public IPs: 44.195.91.150 Private IPs: 10.0.6.25

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16. SS-12-Permissions.png

```
aws [Option+S] Search United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-15 12:58:28 UTC; 18s ago
     Invocation: f05d2e37ef4e41b0bc482b5763187169
   Process: 2150 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 2160 (mysqld)
      Status: "Server is operational"
        Tasks: 35 (limit: 627)
      Memory: 478.6M (peak: 483.8M)
         CPU: 907ms
       CGroup: /system.slice/mysql.service
               └─2160 /usr/sbin/mysqld

Jun 15 12:58:26 ip-10-0-6-25 systemd[1]: Starting mysql.service - MySQL Community Server...
Jun 15 12:58:28 ip-10-0-6-25 systemd[1]: Started mysql.service - MySQL Community Server.
ubuntu@ip-10-0-6-25:~$ sudo adduser esportadmin
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for esportadmin
Enter the new value, or press ENTER for the default
  Full Name []: ragini singh
    Room Number []: 01
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
ubuntu@ip-10-0-6-25:~$ id esportadmin
uid=1001(esportadmin) gid=1001(esportadmin) groups=1001(esportadmin),100(users)
ubuntu@ip-10-0-6-25:~$ touch demo.txt
ubuntu@ip-10-0-6-25:~$ chmod 755 demo.txt
ubuntu@ip-10-0-6-25:~$ ls -l demo.txt
-rwxr-xr-x 1 ubuntu ubuntu 0 Jun 15 13:01 demo.txt
ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer) ×

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17. SS-13-CronJob.png

```
aws
[Option+S] Q Search
United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

ubuntu@ip-10-0-6-25:~$ chmod 755 demo.txt
ubuntu@ip-10-0-6-25:~$ ls -l demo.txt
-rwxr-xr-x 1 ubuntu ubuntu 0 Jun 15 13:01 demo.txt
ubuntu@ip-10-0-6-25:~$ crontab -e
no crontab for ubuntu - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano
 2. /usr/bin/vim.basic <---- easiest
 3. /usr/bin/vim.tiny
 4. /bin/ed
Choose 1-4 [1]: 1
crontab: installing new crontab
ubuntu@ip-10-0-6-25:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow command
0 9 * * * echo "Backup Check"
ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer)

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18. SS-14-Docker-Installed.png

```
aws
[Option+S] Q Search
United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

Preparing to unpack .../5-dnsmasq-base 2.92-1ubuntu0.3_amd64v3.deb ...
Unpacking dnsmasq-base (2.92-1ubuntu0.3) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../6-docker.io 29.1.3-0ubuntu4.1_amd64v3.deb ...
Unpacking docker.io (29.1.3-0ubuntu4.1) ...
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../7-ubuntu-fan 0.12.17_all.deb ...
Unpacking ubuntu-fan (0.12.17) ...
Setting up dnsmasq-base (2.92-1ubuntu0.3) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up dns-root-data (2025080400build1) ...
Setting up bridge-utils (1.7.1-4ubuntu3) ...
Setting up pigz (2.8-1build1) ...
Setting up containerd (2.2.2-0ubuntu1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' -> '/usr/lib/systemd/system/containerd.service'.
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' -> '/usr/lib/systemd/system/ubuntu-fan.service'.
Setting up docker.io (29.1.3-0ubuntu4.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' -> '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' -> '/usr/lib/systemd/system/docker.socket'.
Processing triggers for dbus (1.16.2-0ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl start docker
ubuntu@ip-10-0-6-25:~$ sudo systemctl enable docker
ubuntu@ip-10-0-6-25:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-10-0-6-25:~$
```

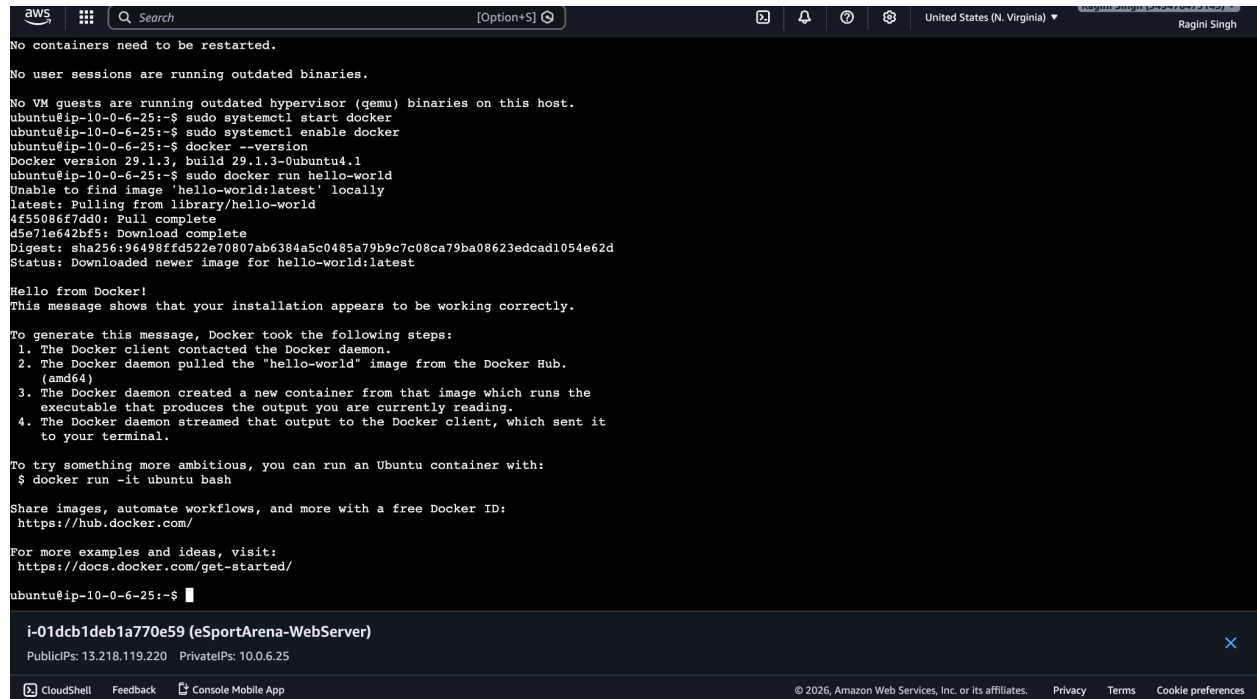
i-01dcb1deb1a770e59 (eSportArena-WebServer)

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19. SS-15-Docker-Container.png



```
aws
[Option+S]
United States (N. Virginia) Ragini Singh

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl start docker
ubuntu@ip-10-0-6-25:~$ sudo systemctl enable docker
ubuntu@ip-10-0-6-25:~$ docker --version
Docker version 20.10.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-10-0-6-25:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer)

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20. SS-16-Shell-Script.png

```
aws [Option+S] United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl start docker
ubuntu@ip-10-0-6-25:~$ sudo systemctl enable docker
ubuntu@ip-10-0-6-25:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-10-0-6-25:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

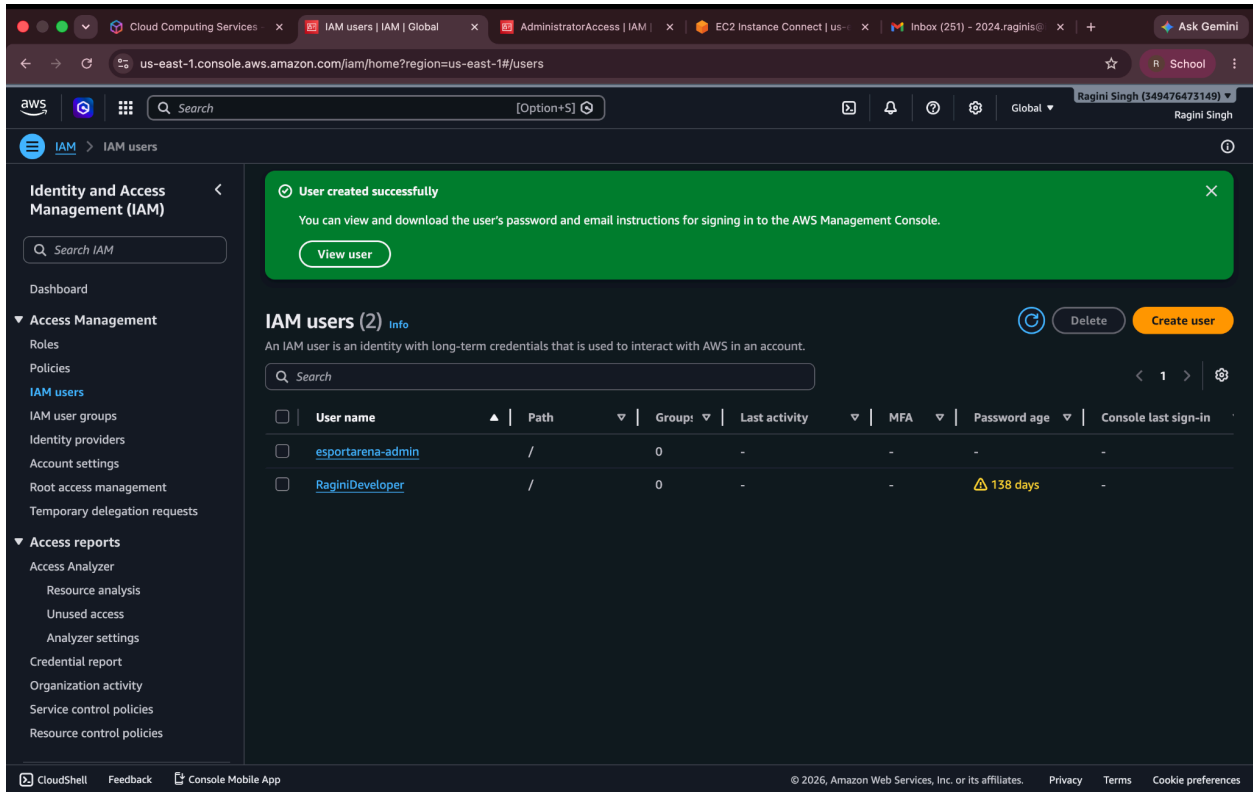
ubuntu@ip-10-0-6-25:~$ nano backup.sh
ubuntu@ip-10-0-6-25:~$ chmod +x backup.sh
ubuntu@ip-10-0-6-25:~$ ./backup.sh
Backup Completed Successfully
ubuntu@ip-10-0-6-25:~$
```

i-01dcb1deb1a770e59 (eSportArena-WebServer) ×
PublicIPs: 13.218.119.220 PrivateIPs: 10.0.6.25

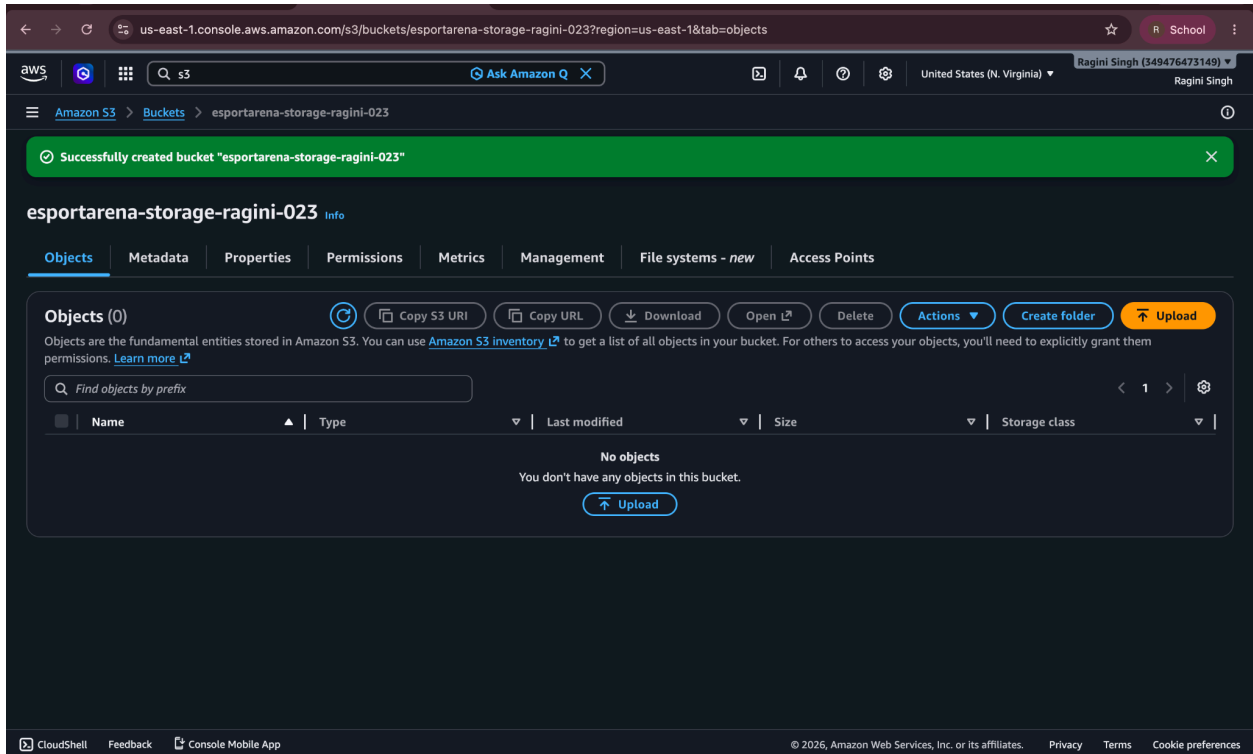
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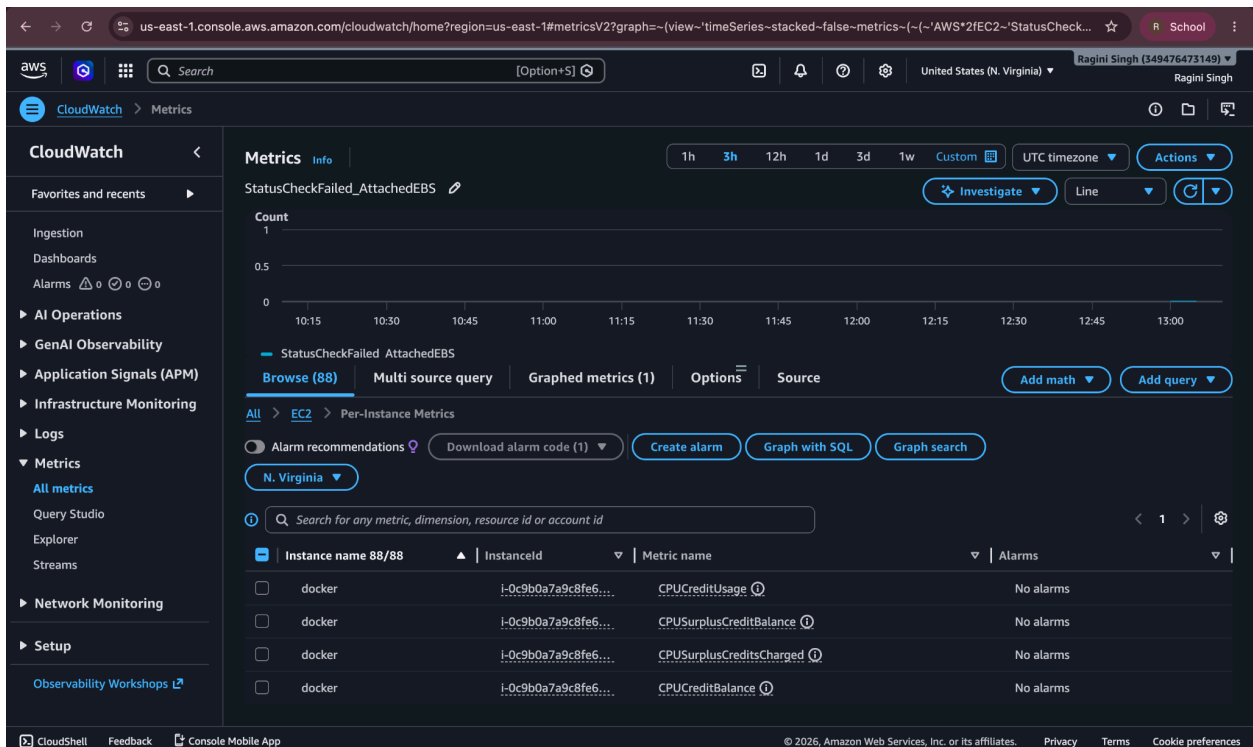
21. SS-17-IAM-User.png



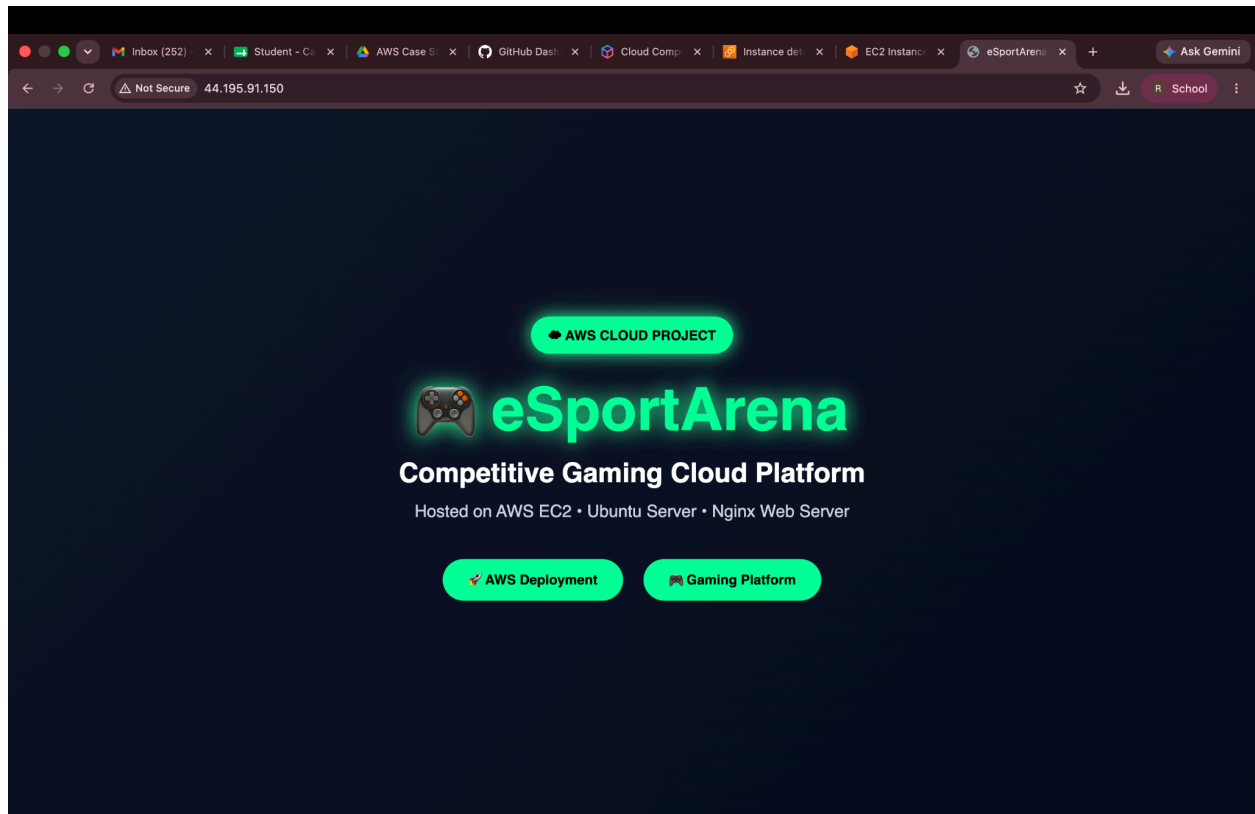
22. SS-18-S3-Bucket.png



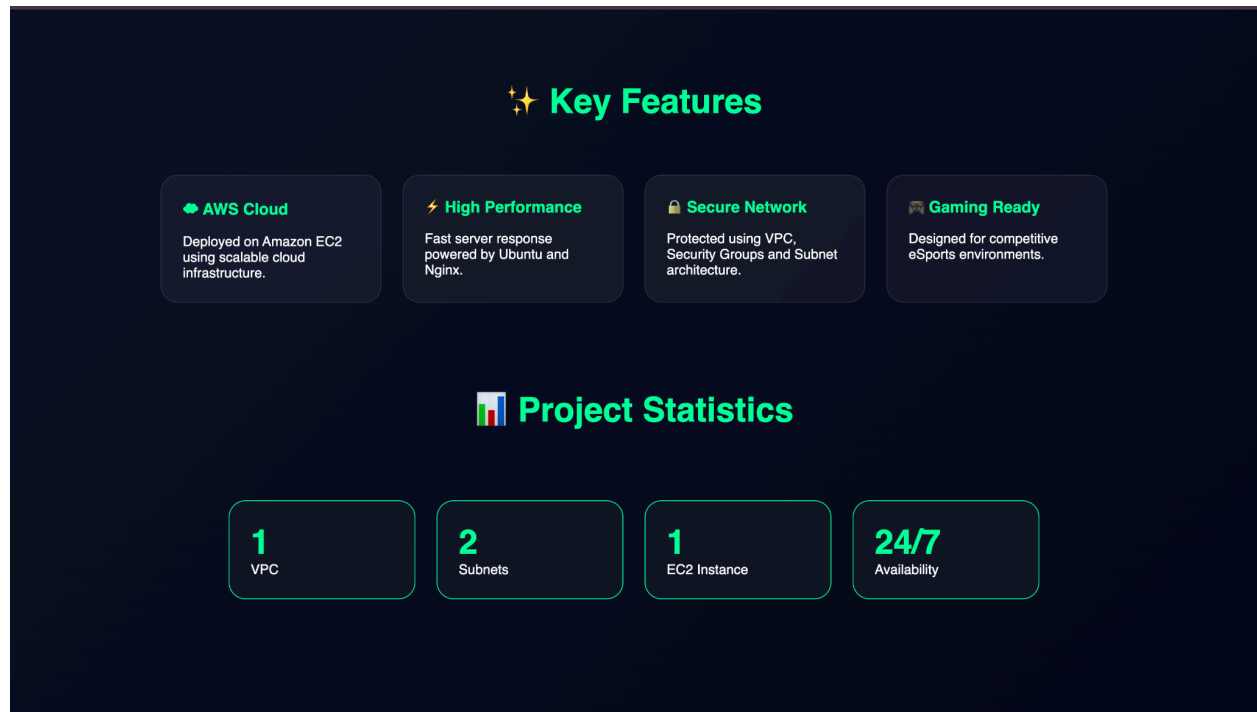
23. SS-19-CloudWatch-Metrics.png



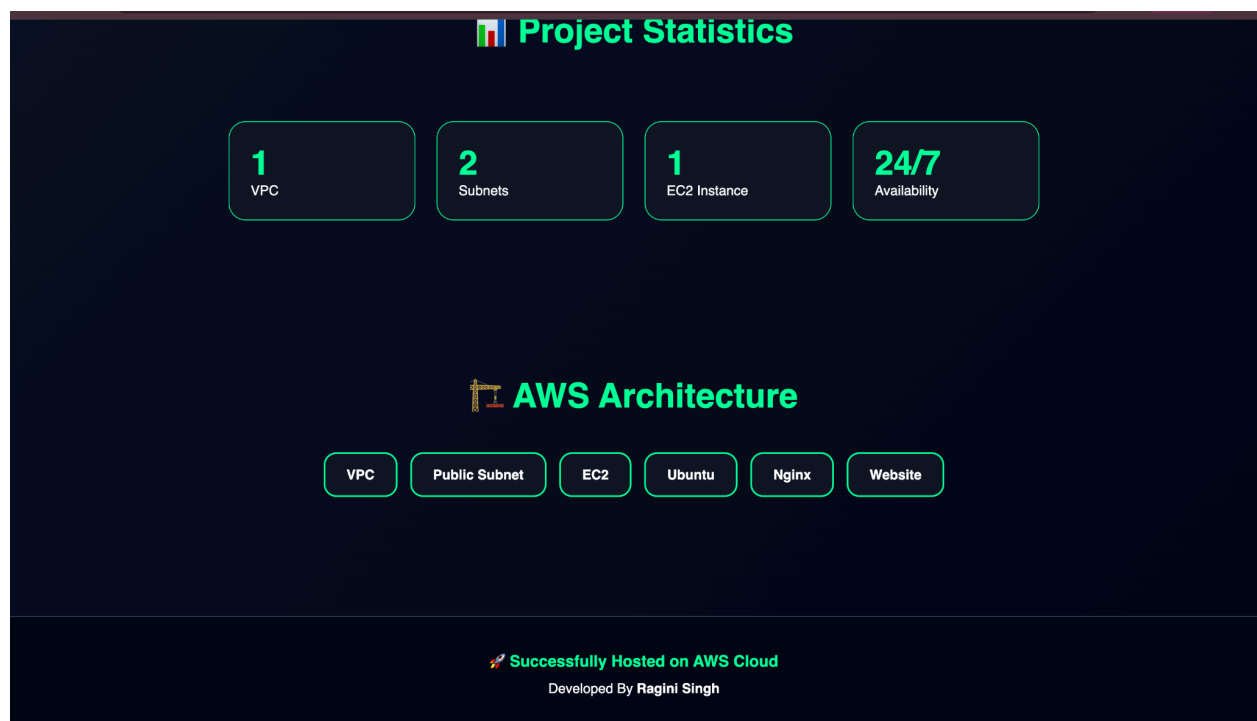
24. Website Home Page.png



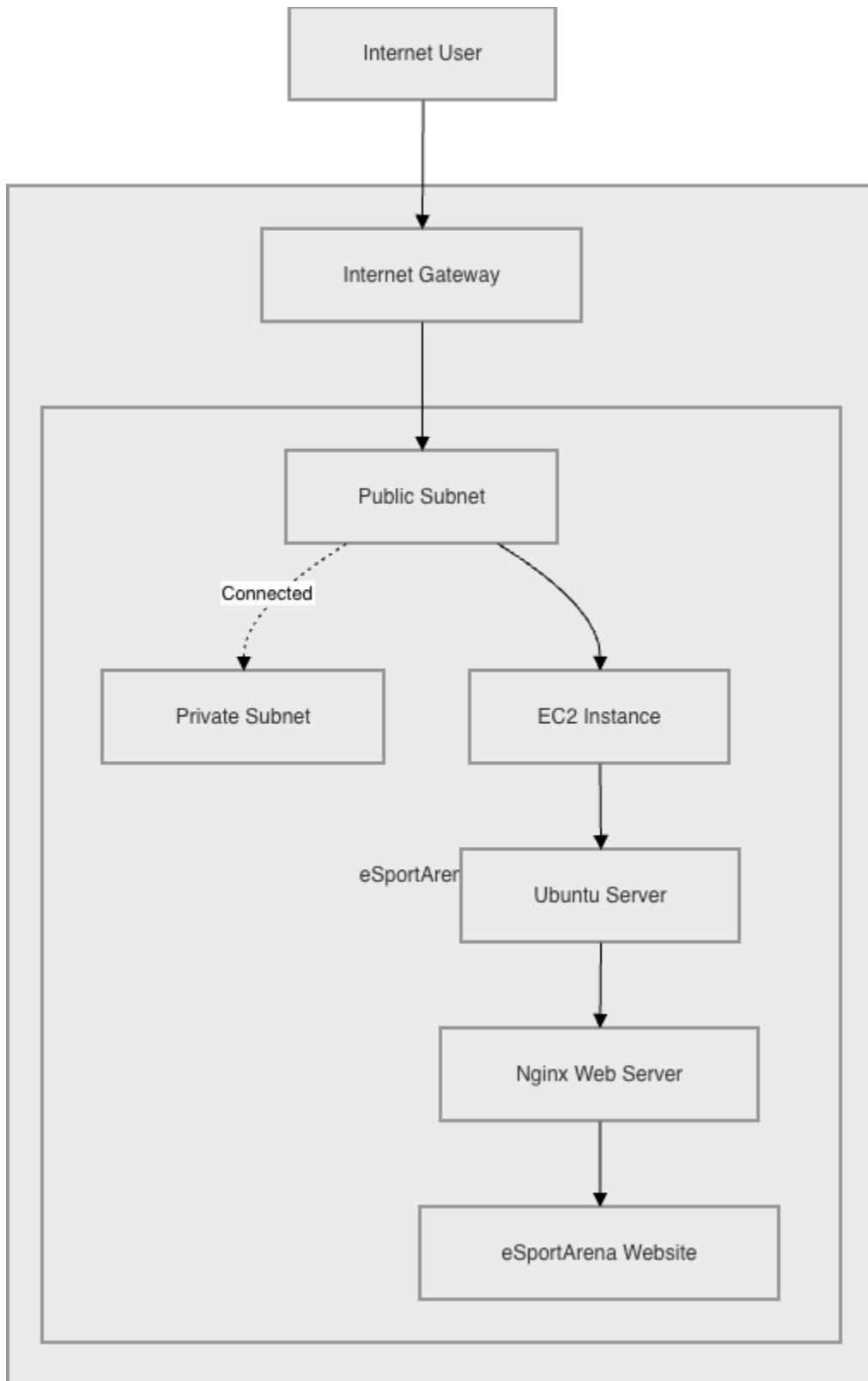
25. Website Features Section.png



26. Website Architecture Section.png



27. AWS_Architecture.drawio.png



7. Results

The eSportArena web application was successfully deployed on AWS EC2 using Ubuntu and Nginx. The website became accessible through the assigned public IP address. The project demonstrates practical implementation of cloud computing concepts including networking, virtualization, security, and web hosting.

8. Future Scope

- Add Load Balancer
- Implement Auto Scaling
- Configure Route 53 Domain
- Deploy HTTPS using SSL Certificates
- Integrate CloudWatch Monitoring
- Containerize Application using Docker
- Deploy using Kubernetes

9. Conclusion

The project successfully demonstrates the deployment of a cloud-hosted gaming platform infrastructure using AWS services. It provides hands-on experience with EC2, VPC, Security Groups, Ubuntu, and Nginx while showcasing the benefits of scalability, availability, and security offered by cloud computing.

THANKYOUUU!!!!!!